Auteur: Siebe Haché
Studierichting: Informatica
Oefeningenreeks 4A nr. 38.9

$$
\begin{aligned}
&\int \frac{x^2+6x+12}{x^2+4x+5}dx \\
&= \int \frac{(x^2+4x+5)+2x+7}{x^2+4x+5}dx \\
&= \int \left(1+\frac{2x+7}{x^2+4x+5}\right) dx \\
&= \int dx + \int \frac{(2x+4)+3}{x^2+4x+5}dx \\
&= \int dx + \int \frac{2x+4}{x^2+4x+5}dx + 3\int \frac{dx}{x^2+4x+5}
\end{aligned}
$$

$$u = x^2+4x+5 \Rightarrow du = (2x+4)dx$$

$$v = x+2 \Rightarrow dv = dx$$

$$
\begin{aligned}
&= \int dx + \int \frac{du}{u} + 3\int \frac{dv}{v^2+1} \\
&= x + \ln|u| + 3\,\mathrm{Bgtan}\, v + C
\end{aligned}
$$

$$= x + \ln|x^2+4x+5| + 3\,\mathrm{Bgtan}(x+2) + C$$

References